ENVIRONMENTAL ENGINEERING

DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, UTTAR PRADESH, LUCKNOW



EVALUATION SCHEME & SYLLABUS

FOR

B. TECH. FOURTH YEAR

ENVIRONMENTAL ENGINEERING

AS PER

AICTE MODEL CURRICULUM

[Effective from the Session: 2021-22]

ENVIRONMENTAL ENGINEERING

SEVENTH SEMESTER ENVIRONMENTAL ENGINEERING SESSION 2021-22

S.No	Subject Code	Subject	Periods			Evaluation Scheme				End Semester		Total	Credit
			L	T	P	CT	TA	Total	PS	TE	PE		
1	KHU701/ KHU702	HSMC-1#/HSMC-2# [Annexure-B(iii)]	3	0	0	30	20	50		100		150	3
2		Departmental Elective-IV	3	0	0	30	20	50		100		150	3
	KNE 071	Construction Technology and Management											
	KNE 072	Engineering Hydrology and Groundwater Management											
	KNE 073	Computer Aided Design of Structure											
	KNE 074	Soil and Water Conservation Engineering											
3		Departmental Elective -V	3	0	0	30	20	50		100		150	3
	KNE 075	Water Power Engineering											
	KNE 076	Integrated Watershed Management											
	KNE 077	Rural Development Engineering											
	KNE 078	Environmental Statistics and Experimental Design											
4		Open Elective – II [Annexure – B(iv)]	3	0	0	30	20	50		100		150	3
5	KNE75X	Elective Lab	0	0	2				25		25	50	1
6	KNE752	Mini Project or Internship Assessment*	0	0	2				50			50	1
7	KNE753	Project I	0	0	8				50		100	150	4
8		MOOCs (Essential for Hons. Degree)											
		Total	12	0	12							850	18

NOTE:

1. **Regular classroom interaction with industry experts is to be ensured in all theory courses (minimum two expert talks from relevant Industry).**
2. **Working on experiments using virtual labs is to be ensured in lab courses.**
3. **Student's visit to Industry/Industry Expert's project site must be arranged as & when possible.**
4. **The Mini Project or Internship (4 - 6 weeks) conducted during semester break after VI semester will be assessed during VII semester.**
5. **Project work is to be identified during VI semester, initiated in VII semester (KNE 753) and completed in VIII semester (KNE 851).**

ENVIRONMENTAL ENGINEERING

EIGHTH SEMESTER ENVIRONMENTAL ENGINEERING SESSION 2021-22

S.No	Subject Code	Subject	Periods			Evaluation Scheme				End Semester		Total	Credit
			L	T	P	CT	TA	Total	PS	TE	PE		
1	KHU801/ KHU802	HSMC-2#/HSMC-1# [Annexure-B(iii)]	3	0	0	30	20	50		100		150	3
2		Open Elective – III	3	0	0	30	20	50		100		150	3
3		Open Elective – IV	3	0	0	30	20	50		100		150	3
4	KNE 851	Project II	0	0	18				100		300	400	9
5		MOOCs (Essential for Hons. Degree)											
		Total	9	0	18							850	18

B. Tech (IV Year) VII Semester

Syllabus

ELECTIVE IV

ENVIRONMENTAL ENGINEERING

KNE-071	Construction Technology and Management	3L:0T:0P	3Credits

Unit	*Topics*	Lectures
I	Elements of Management: Project cycle, Organization, planning, scheduling monitoring updating and management system in construction.	8
II	Network Techniques : Bar charts, milestone charts, work break down structure and reparation of networks. Application of network Techniques like PERT, GERT, CPM AON and AOA in construction management. Project monitoring, cost planning, resource allocation through network techniques. Line of balance technique.	8
III	Engineering Economics: Time value of money, Present economy studies, Equivalence concept, financing of projects, economic comparison present worth method Equivalent annual cost method, discounted cash flow method, analytical criteria for postponing of investment retirement and replacement of asset. Depreciation and break even cost analysis.	8
IV	Contract Management: Legal aspects of contraction, laws related to contracts, land acquisition, labour safety and welfare. Different types of contracts, their relative advantages and disadvantages. Elements of tender preparation, process of tendering pre-qualification of contracts, Evaluation of tenders, contract negotiation and award of work, monitoring of contract extra items, settlements of disputes, arbitration and commissioning of project.	8
V	Equipment Management: Productivity, operational cost, owning and hiring cost and the work motion study. Simulation techniques for resource scheduling. Construction Equipments for earth moving , Hauling Equipments, Hoisting Equipments , Conveying Equipments , Concrete Production Equipments.	8

Text Books

1. Construction Planning, Equipment and Methods. : R.L. Peurify. T.M.H., International Book Company.
2. PERT & CPM Principles and Applications L.S. Srinath, E.W.P. Ltd., New Delhi.
3. Network Analysis Techniques S.K. Bhatnagar, Willey Eastern Ltd.
4. Construction Technology by Sarkar , Oxford
5. Construction Project Management by KK Chitkara, McGraw Hill Publication.
6. Construction Management and Planning by Sengupta and Guha, McGraw Hill

ENVIRONMENTAL ENGINEERING

KNE-072	Engineering Hydrology And Groundwater Management	3L:0T:0P	3Credits

Unit	*Topics*	Lectures
I	Ground Water Occurrence: Ground water hydrologic cycle, origin of ground water, rock properties effecting ground water, vertical distribution of ground water, zone of aeration and zone of saturation, geologic formation as Aquifers, types of aquifers, porosity, Specific yield and Specific retention, Groundwater Basin Management: Concepts of conjunction use, Case studies.	8
II	Ground Water Movement: Permeability, Darcy's law, storage coefficient. Transmissivity, differential equation governing ground water flow in three dimensions derivation, ground water flow equation in polar coordinate system. Ground water flow contours their applications.	8
III	Analysis of Pumping Test Data – I: Steady flow groundwater flow towards a well in confined and unconfined aquifers – Dupit's and Theism's equations, Assumptions, Formation constants, yield of an open well interface and well tests, Analysis of Pumping Test Data – II: Unsteady flow towards a well – Non equilibrium equations – Thesis solution – Jocob and Chow's simplifications, Leak aquifers.	8
IV	Surface and Subsurface Investigation: Surface methods of exploration – Electrical resistivity and Seismic refraction methods. Subsurface methods – Geophysical logging and resistivity logging. Aerial Photogrammetry applications along with Case Studies in Subsurface Investigation.	8
V	Artificial Recharge of Ground Water: Concept of artificial recharge – recharge methods, relative merits, Applications of GIS and Remote Sensing in Artificial Recharge of Ground water along with Case studies, Saline Water Intrusion in aquifer: Occurrence of saline water intrusions, Ghyben Herzberg relation, Shape of interface, control of seawater intrusion.	8

Text Books:

1. Ground water Hydrology by David Keith Todd, John Wiley & Son, New York.
2. Groundwater by H.M.Raghunath, Wiley Eastern Ltd.
3. Groundwater Hydrology by Bhagu R. Chahar, Mc Graw Hill Publication Ltd.

References:

1. Groundwater by Bawvwr, John Wiley & sons.
2. Groundwater Syatem Planning & Managemnet – R.Willes & W.W.G.Yeh, Printice Hall.
3. Applied Hydrogeology by C.W.Fetta, CBS Publishers & Distributers
4. Construction Management and Planning by Sengupta and Guha, McGraw Hill

KNE-073	Computer Aided Design of Structure	3L:0T:0P	3Credits

ENVIRONMENTAL ENGINEERING

Unit	*Topics*	Lectures
I	Elements of Computer Aided Design and its advantages over conventional design. Hardware required for CAD works.	8
II	Principles of software design, concept of modular programming, debugging and testing.	8
III	Computer applications in analysis and design of Civil Engineering systems.	8
IV	Use of software packages in the area of Structural, Geotechnical, and Environmental fields.	8
V	Expert system, their development and applications, Introduction to Neural Networks.	8

Reference:

1. Computer Aided Design – S. Rajiv, Narosa Publication
2. A.I. and Expert System – Robert L. Lerine & / Lane E. Drang, McGraw Hill
3. Neural Computing: Waserman, vonnostrand.
4. Auto CAD 2013, Dummes Bill Fane
5. 5.Cad Frame & Architecture by Pieter Van Der Wolf

ENVIRONMENTAL ENGINEERING

KNE-074	Soil and Water Conservation Engineering	3L:0T:0P	3Credits

Unit	*Topics*	Lectures
I	Definition and scope of soil conservation, cause of soil erosion, Mechanism of erosion, universal soil loss equation, soil erosion due to wind and its control, vegetation management, i.e., strip cropping, stubble mulching and other practices.	8
II	Types of soil erosion due to water- sheet erosion, rill erosion, gully erosion, sediment transport in channels, sediment deposition in reservoirs. Methods of soil erosion control: bounding and terracing on agriculture land for gully control, bench terraces, vegetated water ways, chute spillways, drop inlet spillways, check dams, river training works.	8
III	Biological methods of soil erosion control, grass land management, forest management. Soil quality management, drainage works, reclamation of salt affected soils. Water conservation: water harvesting, rainfall- run off relation, water storage in ponds, lakes, reservoirs and aquifers, groundwater recharge through wells, check dams and storage works.	8
IV	Water losses: filtration, seepage and evaporation losses, pollution/ contamination of water quality due to agricultural practices i.e., fertilizers and pesticides, self purification of surface water, sources of agricultural water pollution, pollutant dispersion in ground water.	8
V	Need of planned utilization of water resources, economics of water resources utilization. Flood plain zones management, modifying the flood, reducing susceptibility to damage, reducing the impact of flooding.	8

Suggested reading:

1. Alam Singh – Modern Geotechnical Engineering
2. K. R. Arora – Soil Mechanics and foundation Engineering.
3. N. C. Brady – Principles of Soil Sciences
4. B. C. Punmia – Soil Mechanics and Foundation Engineering

ELECTIVE V

ENVIRONMENTAL ENGINEERING

KNE-075	Water Power Engineering	3L:0T:0P	3Credits

Unit	*Topics*	Lectures
I	Water Power Introduction: Source of Energy, Status of hydro power in the World. Hydro – Power Place of Hydro Power in a Power system, Transmission Voltages and Hydro-power, estimation of water power potential, General load curve, load factor, capacity factor, utilization factor, diversity factor, load duration curve, firm power, secondary power, prediction of load illustrative examples.	8
II	Type of Hydro-Power Plants –I Classification of Hydel Plants, run of river plants, general arrangement of run of river plants, valley dam plants, diversion canal plants, high head diversion plants storage and pondage illustrative examples. Type of Hydro Power Plants –II Basic features historical development, advantages of pumped storage plants, types of pumped storage plants, relative merits of two unit and three unit arrangement. Three unit arrangement, reversible pump turbines, problems of operation, topography reservoirs and water conveyance, power house, efficiency of P-S plants, illustrative example.	8
III	Water Conveyance General. Classification of penstocks, design criteria for penstocks, economical diameter of penstock, anchor blocks, conduit valves, types of valves, bends and manifolds, illustrative example, Introduction, water hammer, resonance in penstocks, channel surges, surge tanks illustrative examples. Intakes, type of intakes, losses of intakes, air entrainment at intakes, inlet aeration, canals fore bay, tunnels.	8
IV	Turbines Introduction, main types of turbines , hydraulic features, turbine size, constructional features of turbines, layout arrangements, hydraulic of turbines, basic flow equations, draft tubes, cavitations in turbines, governing of turbines, turbine model testing characteristics of turbines, illustrative examples.	8
V	Power House Planning General. (A) surface power stations, power house structure, power house dimensions, lighting and ventilation, variations in design of power house (B) underground power station, history, location of U.G power station, Types of U.G power station, advantages of U.G power house, components of U.G power house, types of layout, limitations of U.G power house structural design of power house, Tidal phenomenon, tidal power- basis principle, historical development, location of tidal power plant, difficulties in tidal power generation, components of tidal power plants, modes of generation, single basin arrangement, double basin system.	8

Suggested reading:

1. Water Power Engineering by M.M. Dandekar and K.N. Sharma, Vani Educational Books
2. Irrigation and water resources Engg. By G.L. Asawa New age international publishers.
3. Irrigation and water power Engineering by B.C. Punamia, Pande B.B. lal (Laxmi Publications Private Limited)
4. Irrigation Water Resources and Water Power Engineering by Dr. P.N. Modi, Standard book House New Delhi.

ENVIRONMENTAL ENGINEERING

KNE-076	Integrated Watershed Management	3L:0T:0P	3Credits

Unit	*Topics*	Lectures
I	INTRODUCTION: Concept of watershed development, objectives of watershed development, need for watershed development in India, Integrated and multidisciplinary approach for watershed management.	8
II	WATER HARVESTING: CHARACTERISTICS OF WATERSHED: size, shape, physiography, slope, climate, drainage, land use, vegetation, geology and soils, hydrology and hydrogeology, socioeconomic characteristics, basic data on watersheds, Rainwater Harvesting, catchment harvesting, harvesting structures, soil moisture conservation, check dams, artificial recharge, farm ponds, percolation tanks.	8
III	PRINCIPLES OF EROSION: Types of erosion, factors affecting erosion, effects of erosion on land fertility and land capability, estimation of soil loss due to erosion, Universal soil loss equation, MEASURES TO CONTROL EROSION: Contour techniques, ploughing, furrowing, trenching, bunding, terracing, gully control, rockfill dams, brushwood dam, Gabion.	8
IV	LAND MANAGEMENT: Land use and Land capability classification, management of forest, agricultural, grassland and wild land. Reclamation of saline and alkaline soils.	8
V	Planning of watershed management activities, peoples participation, preparation of action plan, administrative requirements.	8

Suggested reading:

1. Watershed Management by JVS Murthy, - New Age International Publishers.
2. Water Resource Engineering by R.Awurbs and WP James, - Prentice Hall Publishers.
3. Reference:
4. Land and Water Management by VVN Murthy, - Kalyani Publications.
5. Irrigation and Water Management by D.K.Majumdar, Printice Hall of India.

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KNE-077	Rural Development Engineering	3L:0T:0P	3Credits

Unit	*Topics*	Lectures
I	Rural Development Planning and Concept of Appropriate Technology: Scope; development plans; various approaches to rural development planning; concept of appropriate technology. Rural development programme/ projects.	8
II	Rural Housing: Low cost construction materials for housing; Architectural considerations for individual and group housing; Composite material - ferro-cement & fly ash, autoclaved calcium silicate bricks and soil-stabilized un-burnt brick; Plinth protection of mud walls; design consideration and construction of: non-erodable mud plaster, Water-proof and fire-retardant roof treatment for thatch roofs. Pre-cast stone masonry; rat-trap bond for walls; Panels for roof, ferro-cement flooring / roofing units, Earthquake resistant measures for low cost houses.	8
III	Water Supply and Rural Sanitation: Sources of water. BIS & WHO water standards. Quality, Storage and distribution for rural water supply works; basic design principles of treatment-low cost water treatment technologies; conservation of water; rainwater harvesting; drainage in rural areas, low cost waste disposal systems; septic tank ; Biogas technology; low cost community & individual Garbage disposal systems, Ferro-cement water storage tanks.	8
IV	Low Cost Roads and Transport: Broad categories of Pavement Layers, types of Granular Sub-Bases and Bases, Bituminous Construction, Surface Treatments for roads in rural areas. Soil Stabilization, Lime, LimeFlyash and Cement Treated Course. Crusher-run-Macadam. Use of local materials. Flexible Pavement: Design factors, Basic Principles, Guidelines for Surfacing for Rural Road.	8
V	Low Cost Irrigation: Consideration of low cost irrigation techniques , drip & sprinkler irrigation systems. Watershed and catchments area development - problems and features of watershed management, watershed structures	8

Suggested reading:

1. A.G.Madhov Rao, D.S.Ramachandra Murthy, Appropriate Technologies for low cost Housing Oxfordand IBH Publishing Co. Pvt .Ltd.
2. CBRI, Roorkee, Advances in Building Mat erials and Construction.
3. C. Satyanarayana Murthy, Design of Minor Irrigation and Canal Structures. Wiley Eastern Ltd.,
4. 4.Document on Rural Road Development in India Volume1& 2; Central Road Research Institute, New
5. Delhi.
6. Water supply and sanitary engineering by Rangwala, .Charotar publication
7. Rural Infrastructure by P.Nair, SBS Publication
8. Rural Infrastructure by Samalia Bihari Verma, Gyaneshwar Prasad & Sahib Kumari Singh, Sarup & Sons.
9. Rural Development by Katar Singh, SAGE Publication
10. Information and Communication Technology for Agriculture and rural development by R. Saravanan, New India Publishing agency.

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ENVIRONMENTAL ENGINEERING

KNE-078	Environmental Statistics And Experimental Design	3L:0T:0P	3Credits

Unit	*Topics*	Lectures
I	Stochastic Processes in the Environment: Probability concepts; Conditional probability and Baye s theorem.	8
II	Environmental Data Analysis: Descriptive statistics; Averaging times; Sample size determination; Sampling frequency and duration.	8
III	Measurement uncertainty; Accuracy and precision; Sample and dynamic blanks; Error propagation; Linear least-squares regression.	8
IV	Trend analysis; Non-parametric statistics. Experiment Design and Hypothesis : Testing : Factorial design of experiments; Confidence intervals; Equality of means.	8
V	Test; Analysis of variance (ANOVA); F-test; Significance of factor effects and their interaction.	8

Suggested reading:

1. A.G.Madhov Rao, D.S.Ramachandra Murthy, Appropriate Technologies for low cost Housing Oxfordand IBH Publishing Co. Pvt .Ltd.
2. CBRI, Roorkee, Advances in Building Mat erials and Construction.
3. C. Satyanarayana Murthy, Design of Minor Irrigation and Canal Structures. Wiley Eastern Ltd.,
4. 4.Document on Rural Road Development in India Volume1& 2; Central Road Research Institute, New
5. Delhi.
6. Water supply and sanitary engineering by Rangwala, .Charotar publication
7. Rural Infrastructure by P.Nair, SBS Publication
8. Rural Infrastructure by Samalia Bihari Verma, Gyaneshwar Prasad & Sahib Kumari Singh, Sarup & Sons.
9. Rural Development by Katar Singh, SAGE Publication
10. Information and Communication Technology for Agriculture and rural development by R. Saravanan, New India Publishing agency.

ENVIRONMENTAL ENGINEERING

KNE-751A	Finishing Lab	0L:0T2P	1 Credits

1. Study of the environmental problems in the study area.
2. Sampling work and analysis in the lab.
3. Field study and primary data collection.
4. Secondary data collection from agencies.
5. Statistical analysis of data, model development and estimating pollutant quantities.
6. Designing of system using software/ model/ data.
7. Preparing a map using GIS software and report writing

KNE-751B	Environmental System Modeling Lab	0L:0T2P	1 Credits

1. Exercise on computer simulation of air pollution.
2. Exercise on computer simulation of surface water quality.
3. Exercise on computer simulation of soil water balance.
4. Exercise on application of storm water management model.
5. Exercise on application of linear programming in environmental engineering.
6. Exercise on application of transportation problem in environmental engineering
7. Exercise on application of dynamic programming in environmental engineering.